

Challenge: Standardize Text Case

Consistent text formatting is essential for reliable data analysis and grouping. When text data contains a mix of uppercase, lowercase, or capitalized words, grouping and comparison operations can yield misleading results. For instance, "Apple", "apple", and "APPLE" might all refer to the same value, but without standardization, they are treated as distinct entries. By ensuring that all text values in a column use the same case, you simplify grouping and aggregation, reduce errors, and make your data easier to work with.

```
import pandas as pd

data = {
    "fruit": ["Apple", "banana", "ORANGE", "apple", "Banana",
"orange"],
    "quantity": [5, 3, 4, 2, 1, 6]
}
df = pd.DataFrame(data)
print(df)
```